

ROKIT Healthcare: Building a fast and safe global regenerative treatment platform

Tell us your challenge. We're here to help.

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ABOUT ROKIT



ROKIT Healthcare
Vertex AI to build
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About ROKIT Healthcare

ROKIT Healthcare researches and develops medical organ regeneration solutions with the goal to personalize treatment methods to match the needs of each person's different biological environments and characteristics, even if they suffer from the same diseases. Building on its expertise in regenerative medicine, with artificial intelligence technology, ROKIT is providing the safest, most effective, and fastest regenerative platform anywhere in the world.

Industries: Healthcare

Google Cloud results

- Build fast and reliable AI pipelines with Vertex AI
- Develop and operate machine learning-based technology for the global market with a small number of developers
- Reliable development team that provides support whenever required

Simpl
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Location: Korea

Kubernetes Engine

Google Cloud (<https://cloud.google.com/>)

Vertex AI (<https://cloud.google.com/vertex-ai/>)

Kubernetes Engine

(<https://cloud.google.com/kubernetes-engine/>)

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Our body has a great ability to regenerate and heal all wounds back to their original state. However, the recovery process is more difficult when the wound is too large and deep, or when a disease such as diabetes interferes with regeneration. In these

cases, healing can be improved if the tissue that helps regeneration in our body can be made closer to the wound. A Korean company with technology to make bio patches that help skin and cartilage grow back, ROKIT Healthcare (<https://rokithealthcare.com/>) set out to help wounds heal better with a patch that can be made and attached to a wound by taking tissue directly from the patient's body.

The process starts by making 3D bio-ink by inhaling the fat layer from the abdomen or thighs and filtering out only substances that help regeneration. From this, they are able to print a patch designed to size for the affected area with a dedicated 3D printer. The regenerative patch's safety is enhanced by only using the necessary ingredients from the patient's own body, without adding any additional ingredients. This means there is a reduced risk of side effects.

As every wound is unlike another, a customized patch is required for optimal healing. This 'personalization' of every patch is where ROKIT Healthcare's success lies. ROKIT Healthcare identified the importance of being able to easily make the 3D patches directly in hospitals for the treatment to be most effective. However, it is not easy to perform direct scanning, CAD modeling, and refining and printing of bio-ink in a hospital operating room.

The answer to these challenges was found in the Cloud and with the help of Artificial Intelligence.

Dr. Han Joo Chae, Chief Artificial Intelligence Officer (CAIO) of ROKIT Healthcare, turned to Google Cloud (<https://cloud.google.com/>) for a solution, focusing on machine learning elements and global services.

"Reducing time and increasing accuracy with artificial intelligence was a priority for us so that the technology could be implemented anytime around the world. Google Cloud, which has various artificial intelligence capabilities, was best suited to our needs," he says.

"We chose Google Cloud because we had a goal of getting things up and running quickly and reliably with a small development team. The cloud isn't just infrastructure, it's the services that are provided on top of it, so if there are services that we can rely on, we'll take advantage of them rather than develop them ourselves," adds Dr. Chae.

In particular, Vertex AI (<https://cloud.google.com/vertex-ai>), operated more smoothly than any other service by integrating a series of iterative processes in which a machine learning model was developed, operated, learned again, and a better model was deployed. All the data is stored and processed in Google Cloud in real time, supporting ROKIT Healthcare's goal of providing the same level of medical service anywhere in the world.

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Personalized medical services with Vertex AI

The process of making a 3D bio patch is connected to the data flow of Google Cloud. First, you need to accurately measure the shape of the affected area and capture data on the curvature of the area.

Therefore, a sensor that can read depth as well as a general camera that recognizes the shape of an object with the three RGB colors that make up the light is required. Usually, these cameras are expensive, but ROKIT Healthcare has been able to use sensors on smartphones and tablets. This enables the data produced to be very precise without the high cost.

The data of the affected area is immediately transmitted to the Google Cloud. Google Cloud's Vertex AI then analyzes the data through a machine learning model developed by ROKIT Healthcare, interpreting the shape of the affected area as 3D data and processes it so that the 3D printer can accept it.

Technically, it is not impossible to build in 3D by directly fitting the body, but it takes a long time because it needs a frame to hold the shape, and more bio-ink using cellular tissue is required. ROKIT Healthcare's machine learning model, trained in advance by Vertex AI, interprets the curvature of the affected area and inversely calculates the final

shape so that the 2D patch can be attached in three dimensions. Because the regenerative patch is stretchy, even if the patch looks flat, it adheres exactly to the affected area when applied to the skin. The process of printing a 3D shape into 2D is a key technology that's helped the team to quickly and accurately create a regenerative patch.

All of these processes take place without needing to leave the operating room. The entire procedure only takes 30 minutes. From taking a picture with your tablet and scanning the affected area, it takes only 30 seconds to send it to Vertex AI and create it as actual 3D printing data through ROKIT Healthcare's 3D Vision AI.

Creating an AI-driven data flow, MLOps-driven work environment

The part that Dr. Chae paid most attention to the process of creating an AI pipeline to process artificial intelligence data natively on Google Cloud. The aim was to create an 'MLOps' environment, so that development, operation, and artificial intelligence models run organically.

"Instead of thinking deeply about the detailed elements of the cloud, I started designing using the service I was originally familiar with. However, I came across Google Cloud while deciding to use a Kubernetes container-centered development and operating environment and thinking about how to operate machine learning more safely."

Dr. Chae explains that Vertex AI handles all processes of development, operation, and distribution, which played a big role in bringing ROKIT Healthcare's idea to life. Also, he said that Vertex AI was able to create a single, integrated environment that ties learning, distribution, and services together, not just designing machine learning models, so it could create a seamless picture of the entire solution.

"With the establishment of the Vertex AI-based MLOPS environment, the process of training, deploying, and operating models has been reduced to only a tenth of the work initially envisioned. We were able to focus only on model development, relieving the burden on automation infrastructure and operations from the cumbersome tasks of operation, such as repetitive learning and redeployment," says Dr. Chae.

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ROKIT Healthcare uses [Google Kubernetes Engine](https://cloud.google.com/kubernetes-engine) (<https://cloud.google.com/kubernetes-engine>) to quickly apply various services. Since Vertex AI is built with Kubernetes, it was possible to create a service right away by developing a model in Vertex AI without having to separately build a machine learning operating environment that plays a key role in the design of regenerative patches.

"The most impressive thing is that with the Kubernetes MLOps environment built around Vertex AI, we can focus on model development. Now, the process of training, deploying, and operating a model is only 10% of what was initially thought. The other 90% is automated, so new machine learning models are running smoothly on sites around the world."

Creating a medical experience for the world

Machine learning models produce desired results quickly and precisely, but require considerable effort from developers. The key is to constantly clean the data and experiment with machine learning models to determine the final environment in which to deploy them. In a real environment, a data workflow must be created to facilitate this.

"With Google Cloud, all data from model operation to new experiment results is gathered in one place and seamlessly connected to each other, so there is no technical burden even when hundreds of experiments are conducted. Google Cloud has exceeded our expectations as we've been able to

operate three times better than expected when it comes to infrastructure and development."

ROKIT Healthcare is recently expanding the scope of bio patch to regenerate cartilage as well as skin. Since cartilage can also help regenerate by using the patient's own tissues in a similar way to skin, self-regeneration procedures can be performed in a similar process.

"Since cartilage is not exposed like skin, a new machine learning model is needed to see and judge wounds. This, too, was able to expand the scope by creating and learning a new Vertex AI model according to cartilage regeneration. Once the model is completed, it is distributed to the medical field and the process of creating a regenerative patch is the same."

Since everything is automated and Google Cloud services are available anywhere in the world with high speed and stability, ROKIT Healthcare's 3D patch treatment has the potential to be global, in line with its vision to reduce medical barriers.

"The goal is to ensure that patients can have the same experience no matter who performs the procedure, anywhere in the world. We created an easy-to-use kit and established a global cloud environment so that doctors who can provide skin-related treatment can achieve the same results with just a tablet and 3D printer, even if they are not specialists. I want to show that skin and cartilage can be regenerated without difficulty even in

countries with poor medical conditions," says Dr. Chae.

He also noticed the alignment in values and philosophies between the two businesses. "Google Cloud is not just a company that provides cloud services, but it gives us a sense of stability and expectations from the fact that we have the same goal." After all, technology does not end with technology itself, but its real value is in how it can help to change the world and find groundbreaking solutions.

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